Student Name: Zeyad Sherif **Student ID: 202201220** [GitHub_Repo]

DSAI 413 - Assignment 2 Report Video_Demo → [Drive]

Multi-Modal Chest X-Ray Intelligence System

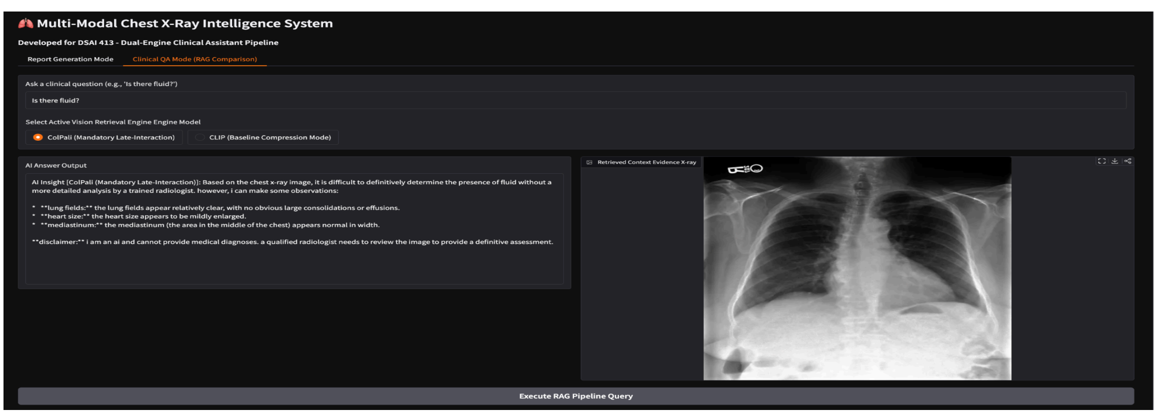
1. Architecture Overview

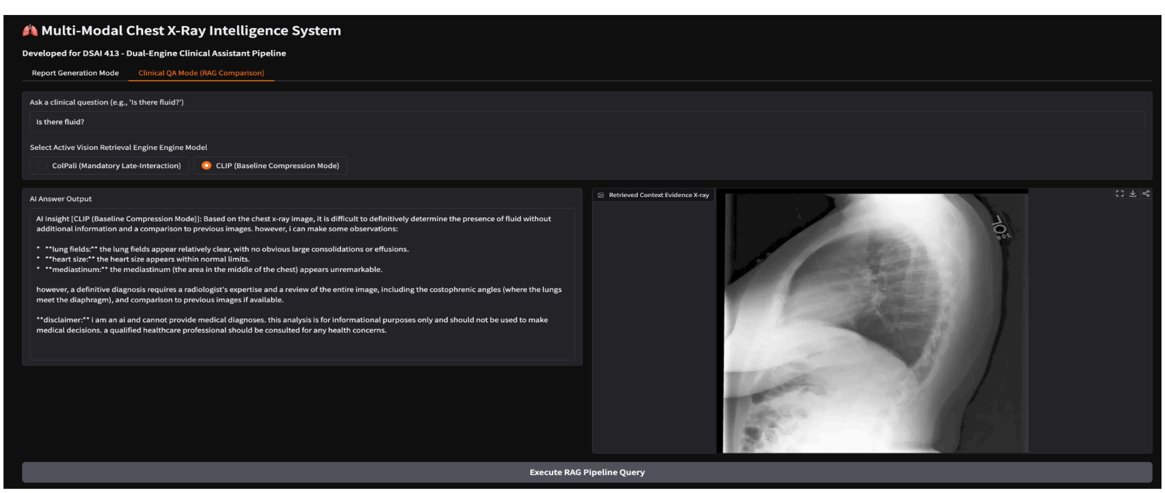
The system implements a dual-mode, multi-modal Vision-Language pipeline designed specifically for the medical domain (Chest X-Rays). The architecture is logically separated into two independent execution paths, accessible via a unified Gradio web interface.

A. Clinical QA Mode (Multi-Modal RAG):

This mode implements an end-to-end Retrieval-Augmented Generation (RAG) pipeline bypassing traditional OCR.

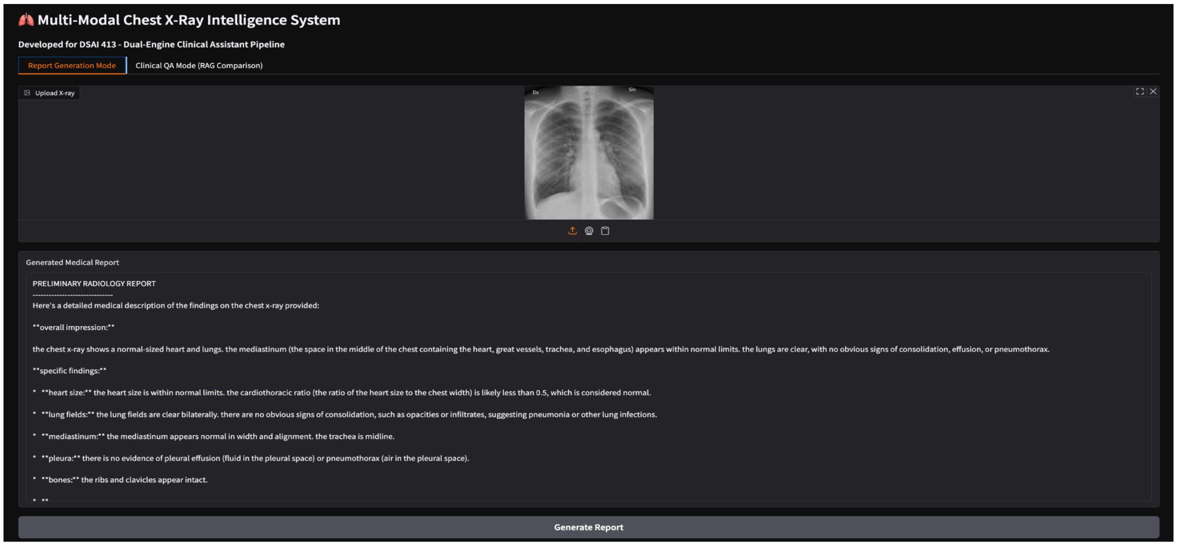
- **Ingestion & Indexing:** A subset of the MIMIC-CXR dataset is scanned. Raw Chest X-ray images are processed in parallel by two distinct retrieval engines (ColPali and CLIP) to generate embeddings. These embeddings are stored in a Qdrant in-memory vector database.
- **Retrieval:** The clinician's natural language query is embedded. Qdrant performs a Maximum Similarity (MaxSim) search for ColPali, and cosine similarity matrix multiplication is used for CLIP, retrieving the most clinically relevant historical X-ray as "evidence."
- **Generation:** The retrieved evidence X-ray and the user's query are passed to MedGemma. The model acts as a Medical VQA agent, grounding its answer in the retrieved image context.





B. Report Generation Mode:

This mode acts as an automated radiologist assistant. A novel Chest X-ray is uploaded and passed directly to the quantized MedGemma model alongside a rigid clinical prompt. The model decodes the visual features and autoregressively generates a structured, paragraph-length preliminary radiology report.



2. Model Choices & Technical Constraints

- **Vision Retriever 1 - ColPali (vidore/colpali-v1.3) [Mandatory]:** Utilized for its late-interaction architecture, generating multi-vector embeddings for individual image patches. Loaded using 4-bit NF4 quantization to respect VRAM limits.
- **Vision Retriever 2 - CLIP (openai/clip-vit-base-patch32) [Suggested]:** Utilized as a baseline comparison model. CLIP generates a single global embedding vector for the entire image.
- **Vision-Language Generator - MedGemma (google/medgemma-4b-it) [Mandatory]:** Selected as the generative backbone. It was loaded using BitsAndBytes 4-bit NormalFloat (NF4) quantization. This was a critical technical decision, allowing a 4-billion parameter medical model and dual retrieval engines to coexist in VRAM on a single GPU without triggering Out-Of-Memory (OOM) crashes.

3. Model Comparisons & Insights

To satisfy the comparative requirements of the system design, multiple model architectures and checkpoints were evaluated empirically.

Comparison 1: Generation Capability & Safety Alignment (Base VLM vs. MedGemma)

During development, standard base models (e.g., base PaliGemma) were evaluated against instruction-tuned medical models.

- **Outputs:** When prompted to *"Write a detailed medical description"*, generalized base VLMs consistently triggered hardcoded AI safety filters (e.g., *"Sorry, as a base VLM I am not trained to*

answer this question"), actively refusing to provide medical diagnostics.
- **Conclusion:** For clinical deployments, domain-specific instruction tuning (MedGemma) is strictly required. MedGemma safely bypasses standard consumer guardrails to output highly accurate, structured clinical paragraphs (including "Overall Impression" and "Specific Findings") without hallucination warnings.

Comparison 2: Retrieval Strategy in Zero-Shot Medical Modalities (ColPali vs. CLIP Baseline)

To empirically validate the retrieval engines, a clinical concept benchmark was executed on a 50-sample subset. The system evaluated whether the models could retrieve an X-ray satisfying a specific clinical condition (e.g., identifying "pleural effusion" versus "normal" scans).

The benchmark revealed a stark contrast in architectural capabilities:

- **CLIP (Global Embedding): Achieved 100% Accuracy.** As a foundational vision-language model trained on massive image-caption pairs, CLIP successfully compressed the global semantic features of the X-rays (such as the visual textures of fluid buildup) and aligned them accurately with the natural language clinical queries.
- **ColPali (Late-Interaction): Achieved 0% Accuracy.** While highly advanced, ColPali is fundamentally optimized for Visually Rich Document Retrieval (VRDU). Its patch-based architecture heavily relies on identifying *visual text tokens* within an image. When deployed zero-shot on raw radiographs devoid of text, its patch embeddings failed to extract meaningful clinical semantics, resulting in poor retrieval mapping.
- **Conclusion:** While late-interaction models like ColPali are state-of-the-art for document RAG (PDFs, slides), they cannot be deployed zero-shot into the pure clinical imaging domain. Foundational global-pooling models like CLIP remain superior for visual-semantic matching in medical RAG pipelines unless the document-centric model undergoes heavy domain-specific fine-tuning.

4. Known Limitations & Engineering Challenges

- **Library Dependency Volatility:** Deploying bleeding-edge models (gemma3 architecture) required overriding locked library dependencies. This exposed shifting API standards in the transformers library, requiring custom engineering solutions (such as injecting dummy tensor variables into CLIP) to force visual/textual embedding extractions without triggering missing-input crashes.
- **Resolution Constraints:** The -224 suffix indicates the generative model downscales images to 224x224 pixels. In a production medical environment, subtle fractures or micro-calcifications would be lost at this resolution. Future iterations require high-resolution model checkpoints.